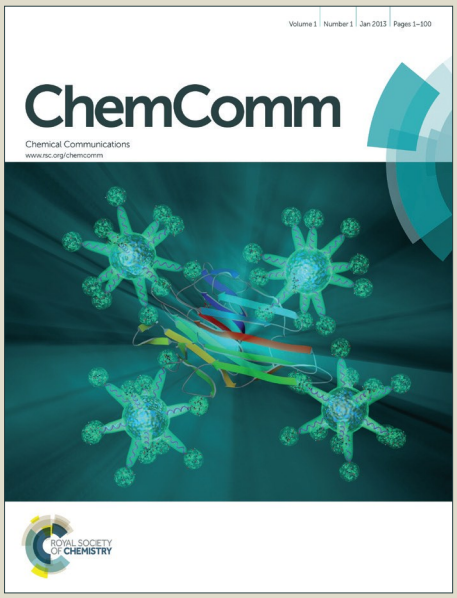
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$CuBr_2$-Catalyzed General Highly Enantioselective Approach to Optically Active Allenols from Terminal Alkynols

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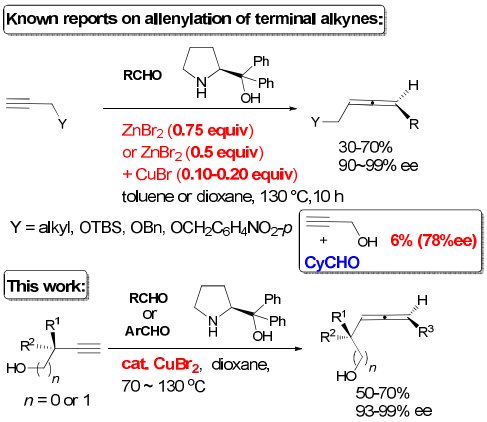
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Xin Huang,[a] Tao Cao,[b,†] Yulin Han,[b,†] Xingguo Jiang,[b†] Weilong Lin,[b,†] Jiasheng Zhang,[b,†] and Shengming Ma[a,b]*

Here we show a $CuBr_2$-catalyzed approach for highly enantioselective synthesis (93~99% ee) of allenols from aldehydes and terminal alkynols with the absolute configuration being controlled by applying the readily available (*R*)- or (*S*)-α,α-diphenylprolinol.

Allenes are unique unsaturated hydrocarbons in comparison to alkenes and alkynes due to the diene-based axial chirality, which was found in over 150 biologically acitive natural compounds and marketed drugs.[1] Furthermore, such axial chirality could be transferred to central chirality via chirality transfer strategy to afford a unique irreplaceable entry to chiral molecules vital for medicine and chemistry provided that the most common functionalities such as alcohol, amide, carboxylate, and malonate may be installed.[2] To fully realize such advantages both in chemistry and medicine for allenes, highly enantioselective efficient synthesis of allenols in the (*R*)- or (*S*)-configuration is of great importance.[3-6]

As we know alcohols are extremely versatile due to their rich reactivities for the syntheses of the corresponding aldehydes, carboxylic acids (esters), tosylates, halides, malonates, amines, amides epoxides, furans, etc. However, chiral allenols are not yet readily available requiring 3 or 5 steps by following the reported methods.[7,8,9c] Recently, *non-catalytic* ZnX_2-mediated[8,9a-c] (in some cases together with CuBr)[9e] allenylation reaction of simple terminal alkynes or propargylic ethers and aldehydes with α,α-diphenylprolinol providing entries to chiral allenes have been developed. However, when *primary* terminal alkynyl alcohols were used under these reported reaction parameters, the results are rather poor[10] (Scheme 1). For non-primary terminal propargylic alcohols, we may prepare the optically active propargylic amines first with CuBr using N-PINAP as the chiral ligand, which were then converted to allenols with Zn^{2+} or Cd^{2+}.[9a,11] In some cases the reactions of aromatic aldehydes failed.[9c] Thus, it is of high interest to develop a new catalyst for such a transformation directly starting from terminal alkynols. Here, we wish to present a versatile direct highly enantioselective approach (93~99% ee) to chiral allenols starting from commonly available starting materials, i.e., terminal alkynols and aldehydes (aliphatic as well as aromatic aldehydes), in which $CuBr_2$ has been identified for the first time as the catalyst for such a tranformation. It is interesting to observe the difference between copper halides with different oxidtion state, i.e., CuBr, which worked efficiently together $ZnBr_2$[9e] and $CuBr_2$!



Scheme 1. Allenylations of terminal alkynes with α,α-diphenylprolinol

After systematic screening, we observed for the first time that by applying 20 mol% $CuBr_2$ as the catalyst, the reaction of propargyl alcohol with *n*-octanal in the presence of the readily available amino alcohol (*S*)-**3** afforded allene (R_a)-**4ja** in 83% ee (entry 1, Table 1). Surprisingly, by simply increasing the ratio of alkynol and aldehyde, the yield improved to 60% and the ee reached 95% (entry 3, Table 1)! We tentatively reasoned that the increased substrates may act as ligand to improve the ee. Increasing the loading of $CuBr_2$ led to lower yield and ee (entries 4 and 5, Table 1). As a comparison, the reaction using CuBr afforded (R_a)-**4ja** in 48% yield with 98% ee (entry 6, Table 1).

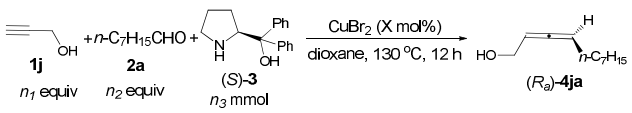
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Table 1.The reaction of propargy alcohol **1j** with aldehyde **2a** in the presence of prolinol (*S*)-**3**.

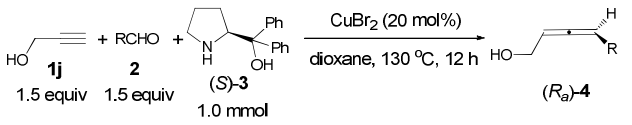


Entry	$n_1/n_2/n_3$	X (mol)	(R_a)-**4ja**	
			Isolated Yield (%)	ee (%)
1[a]	1/1.4/1.2	20	39	83
2	1.2/1.2/1	20	40	92
3	**1.5/1.5/1**	**20**	**60**	**95**
4	1.5/1.5/1	30	51	91
5	1.5/1.5/1	40	46	71
6[b]	1.5/1.5/1	20	48	98

[a] The reaction time was 16 h. [b] CuBr was used instead of $CuBr_2$.

By following this optimized protocol (entry 3, Table 1), the reactions of a wide range of normal or secondary alkyl aldehydes afforded the corresponding axially chiral primary α-allenols in 51~70% yields with 93~98% ee at 70 °C (with much higher 98% ee) or under reflux (for shorter reaction time) (entries 1-16, Table 2). This reaction also worked with aromatic aldehydes substituted with either an electron-withdrawing or electron-donating group, affording the axially chiral primary aryl-substituted α-allenols in 53~56% yields with 95~98% ee also at 70 °C (entries 17-20, Table 2). Such a reaction could be easily carried out on a 50 mmol scale (entry 6, Table 2).

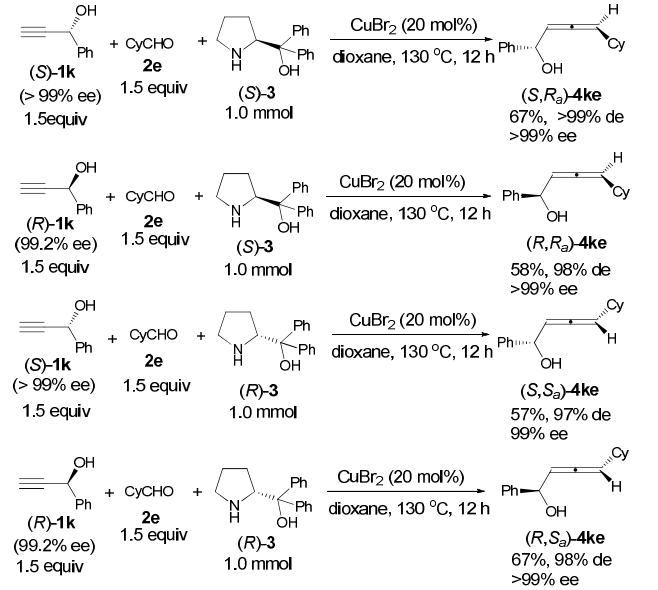
Table 2.The reaction of propargyl alcohol **1j** with aldehydes **2** and prolinol (*S*)-**3**.



Entry	**2**	(R_a)-**4**
	R	Isolated Yield (%) (ee (%))
1	n-C_3H_7 (**2h**)	55 ((R_a)-**4jh**) (94)
2[a]	n-C_5H_{11} (**2i**)	61 ((R_a)-**4ji**) (95)
3	n-C_6H_{13} (**2j**)	59 ((R_a)-**4jj**) (94)
4	n-C_7H_{15} (**2a**)	60((R_a)-**4ja**) (95)
5[a]	n-C_7H_{15} (**2a**)	65 ((R_a)-**4ja**) (95)
6[a,b]	n-C_8H_{17} (**2k**)	60 ((R_a)-**4jk**) (95)
7[c]	n-C_8H_{17} (**2k**)	54 ((R_a)-**4jk**) (98)
8[a]	n-C_9H_{19} (**2l**)	68 ((R_a)-**4jl**) (95)
9[c]	n-C_9H_{19} (**2l**)	54 ((R_a)-**4jl**) (98)
10[a]	n-$C_{10}H_{21}$ (**2m**)	70 ((R_a)-**4jm**) (95)
11[c]	n-$C_{10}H_{21}$(**2m**)	51 ((R_a)-**4jm**) (98)
12	n-$C_{11}H_{23}$ (**2n**)	60 ((R_a)-**4jn**) (93)
13[c]	n-$C_{11}H_{23}$ (**2n**)	54 ((R_a)-**4jn**) (98)
14	$PhCH_2CH_2$ (**2c**)	61 ((R_a)-**4jc**) (95)
15[a]	*i*-Bu (**2d**)	58 ((R_a)-**4jd**) (97)
16	Cy (**2e**)	62 ((R_a)-**4je**) (94)
17[c,d]	*p*-BrC_6H_4 (**2o**)	56 ((R_a)-**4jo**) (98)
18[c,d,e]	*m*-NCC_6H_4(**2p**)	53 ((R_a)-**4jp**) (98)
19[c,d,e]	*p*-MeC_6H_4 (**2q**)	55 ((R_a)-**4jq**) (95)
20[a,c,d]	*p*-PhC_6H_4 (**2r**)	55 ((R_a)-**4jr**) (97)

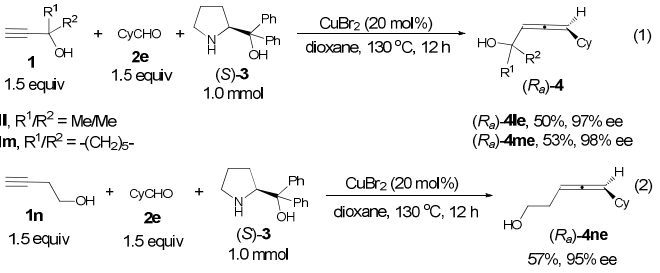
[a] Two equiv each of propargyl alcohol and aldehyde were applied. [b] The reaction was conducted at 50 mmol scale. [c] The reaction was conducted at 70 °C for 24 h. [d] 40 mol% of $CuBr_2$ was used. [e]The product was isolated via column chromatography at 0 °C to avoid racemization.

The highly stereoselective synthesis of axially chiral allenols with extra central chirality is known to be very challenging. Excitingly, the current method also worked with such alcohols bearing an extra central chirality and produced all four stereoisomers (S,R_a)-, (R,R_a)-, (S,S_a)-, and (R,S_a)-**4ke** with an excellent stereoselectivity at one's will with (*R*)- or (*S*)-**3** determing the absolute configuration of the allene unit formed (Scheme 2).



Scheme 2. Diastereoselective Reaction of (*S*)- or (*R*)-1-phenylpropargyl alcohols **1k** with CyCHO.

This protocol is also suitable for the synthesis of chiral 2,3-allenyl tertiary alcohols (R_a)-**4le** and (R_a)-**4me** without decomposing the unstable tertiary alcohol unit (eq. 1 in Scheme 3). Even chiral β-allenol (R_a)-**4ne** could be obtained in 57% yield with 95% ee, indicating the length of the tether linking the hydroxyl group with the allene unit is not critical (eq. 2 in Scheme 3).



Scheme 3. Synthesis of chiral 2,3-allenyl tertiary alcohols (R_a)-**4le**, (R_a)-**4me** and 3,4-allenol (R_a)-**4ne**.

The intermediacy of the propargylic amine (*S,S*)-**5ja** was confirmed by its isolation (eq. 1, Scheme 4) and its treatment with 20 mol% $CuBr_2$ affording allene (R_a)-**4ja** in 53% yield (eq. 2, Scheme 4). Deuterium-labelling experiment applying the deuterated amino alcohol **(*S*)-3-D_2** confirmed the 1,5-D transfer from the 5-position of **(*S*)-3-D_2** to the 2-position of (R_a)-**4ja-D**

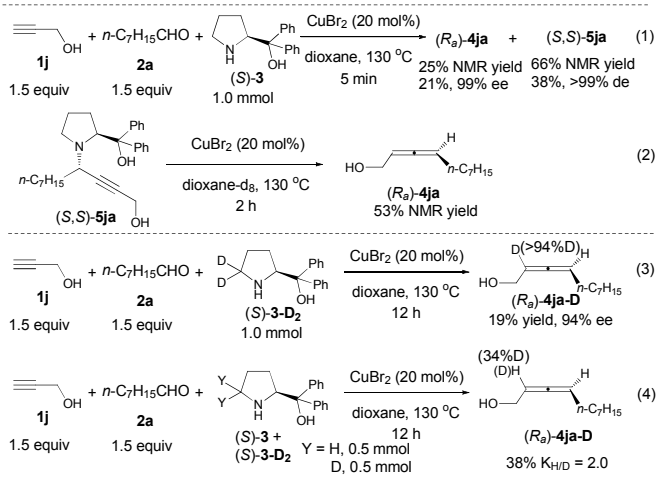
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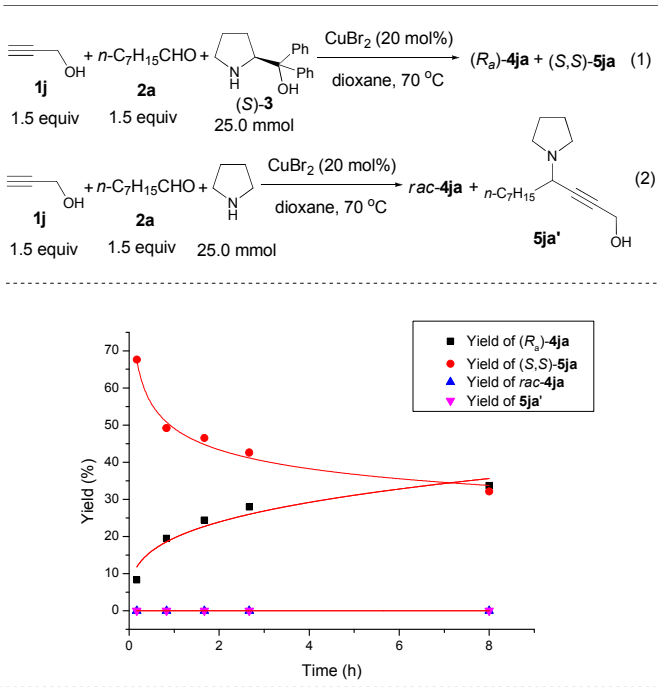
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with a $K_{H/D}$ of 2.0, indicating its rate-determining nature (eqs. 3 and 4, Scheme 4).



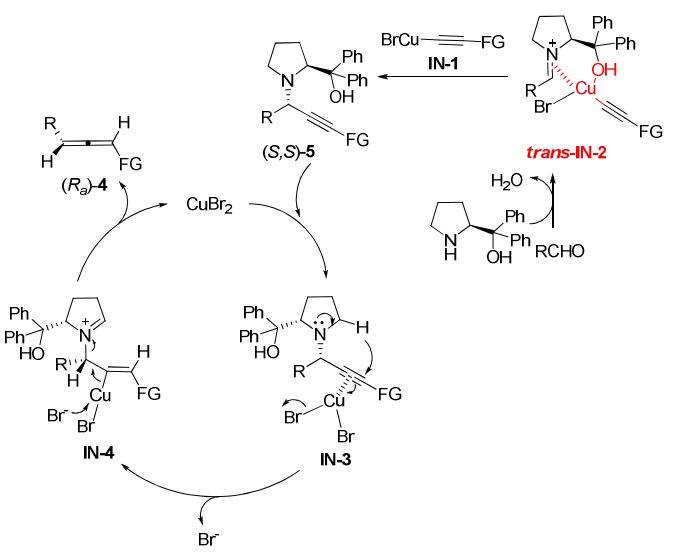
Scheme 4. Control experiments and deuterium-labelling experiments

Surprisingly, the reaction with tetrahydropyrrole without the hydroxyldiphenylmethyl group is extremely sluggish (Scheme 5).



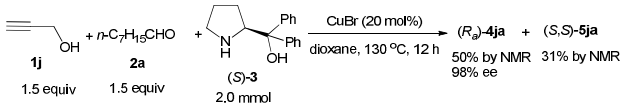
Scheme 5. The effect of amine: α,α-diphenylprolinol vs. tetrahydropyrrole and mechanism

Based on these results, we proposed that the alkynylmetal species **IN-1** would react with the *in situ* generated iminiumion resulting in the smooth 1,2-attack of the alkynyl entity from the back-side of the diphenylhydroxymethyl group through the *hydroxyl*-coordinated tetra-coordinated complex ***trans*-IN-2.** The resulting propargylic amine intermediate (*S*,*S*)-**5** underwent highly stereoselective $CuBr_2$-mediated intramolecular 1,5-hydride transfer via **In-3** and **In-4** followed by β-elimination to afford the (*R*)-allene unit (Scheme 6). Notably, a stoichiometric amount of water generated during the first step did not affect the performance of the Cu^{2+} catalyst.



Scheme 6. The catalytic cycle

Control experiment with CuBr (20 mol%) under the standard reaction conditions showed that the second step reaction of converting the propargylic amine to allene is rather slow (Scheme 7).



Scheme 7. Control experiments with CuBr

In conclusion, the robustness of $CuBr_2$ catalyst for this highly enantioselective approach to 1,3-disubstituted allenols is noteworthy. Simple non-noble divalent $CuBr_2$ is fully compatible with the in-situ generated water, the hydroxyl group, and the cyclic imine formed after β–elimination, which makes it catalytic in this type of transformation. The versatile transformation took advantage of generic starting chemicals-aldehydes and terminal alkynols-commercially available from all of chemical reagents catalogues and controlled the absolute configuration of the allene unit by simply applying the readily available (*R*)- or (*S*)-α,α-diphenylprolinol.[12] Due to the excellent ees, the broad scope, and the potential of these chiral allenols, this versatile yet simple solution will be useful in organic synthesis.[13] Further studies including the scope, synthetic application, and the effect of the ratio of the substrates on the ee are being actively pursued in this laboratory.

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Notes and references

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[†] These authors contributed equally to this work.

Electronic Supplementary Information (ESI) available: experimental section,characterization of all compounds, and copies of ^{1}H and ^{13}C NMR spectra ofselected compounds. See DOI: 10.1039/b000000x/

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10. For $ZnBr_2$-mediated synthesis of non-primary chiral 2,3-allenols from propargylic amines, see also: ref. 9a. Propargyl alcohol also failed in this protocol.
11. For CdI_2-mediated synthesis of non-primary chiral 2,3-allenols from propargylic amines, see also: J. Zhang, J. Ye, S. Ma, *Org. Biomol. Chem.*, 2015, **13**, DOI: 10.1039/C4OB02673J. Propargyl alcohol also failed in this protocol.
12. The prices for (*S*) and (*R*)-α,α-diphenylprolinol are 635 $/kg and 2443 $/kg, respectively, from Shanghai Darui Fine Chemicals.
13. **Preparation of (R_a)-dodeca-2,3-dien-1-ol (R_a)-4jk.** To a flame-dried three-neck flask with a reflux condenser were added $CuBr_2$ (2.2406 g, 10.0 mmol), (*S*)-**3** (12.6504 g, 50.0 mmol), **1j** (5.6047 g, 100.0 mmol)/dioxane (25 mL), and **2k** (14.2570 g, 100.0 mmol)/dioxane (10 mL) sequentially under nitrogen atmosphere. The reaction was complete after being stirred in an oil bath preheated at 130 °C for 12 h as monitored by TLC (eluent: petroleum ether/ethyl acetate = 5/1). After cooling to room temperature, the resulting mixture was diluted with ether (200 mL), and washed with an aqueous solution of hydrochloric acid (3 M, 200 mL). The organic layer was separated, and the aqueous layer was extracted with ether (200 mL × 2). The combined organic layer was washed with brine and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford afforded (R_a)-**4jk** (5.4270 g, 60%) (eluent: petroleum ether/ethyl acetate = 20/1 to 10/1) as a liquid: 95% ee (HPLC conditions: Chiralcel As-H column, hexane/*i*-PrOH = 200/1, 0.6 mL/min, λ = 214 nm, t_R(major) = 24.0 min, t_R(minor) = 25.6 min); $[\alpha]_D^{26.1}$ = -61.7 (*c* = 0.98, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$) δ 5.40-5.20 (m, 2 H, CH=C=CH), 4.21-4.02 (m, 2 H, OCH_2), 2.09-1.92 (m, 2 H, CH_2), 1.69 (br s, 1 H, OH), 1.49-1.12 (m, 12 H, CH_2 × 6), 0.88 (t, *J* = 6.6 Hz, 3 H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 203.0, 93.8, 91.6, 60.7, 31.8, 29.3, 29.2, 29.1, 29.0, 28.6, 22.6, 14.0; IR (neat) ν (cm^{-1}) 3328, 2924, 2856, 1964, 1462, 1378, 1263, 1211, 1057, 1014; MS (EI) m/z 182 (M^+, 0.52), 55 (100); HRMS calcd. for $C_{12}H_{22}O$ [M^+]: 182.1671, Found:182.1667.

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